Benchmarking and Assessing Transformations Between Text and Causal Maps via Large Language Models

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Abstract. Causal maps are specialized ontologies in which concept nodes are connected through typed, directed edges that encode positive or negative causality. These maps can be used to elicit the mental models of participants, thus supporting tasks such as the identification of meaningful groups or the synthesis of comprehensive models of a domain. Although producing causal maps involves a transparent process, the large maps produced by groups are notoriously difficult to interpret. In addition, creating maps is a time-consuming process that requires trained facilitators. These limitations have fueled the interest in automatically explaining maps by transforming them into accessible narratives (i.e., map-to-text) or in creating maps using authoritative reports (i.e., text-to-map). In this brief ontology report, we provide a set of open resources on standard formats to support both tasks. Specifically, we provide 5 datasets that can support map-to-text or text-to-map tasks at different levels (e.g., sentence- or paragraph-level generation), across application domains (e.g., ecological management, public health), and with a variety of writing styles (novice, advanced, experts). We detail assessment procedures for these tasks, covering both existing metrics and emerging approaches. Finally, we provide five notebooks to support users in performing these tasks and assessments through our open datasets.

Keywords: Conceptual Model, Model-to-text, Natural Language Generation, Participatory Modeling, Text-to-model

1. Introduction

Causal maps are a specialized form of ontologies, consisting of labeled nodes and *typed* directed edges. Edges encode causal relationships, thus they are either increasing (+) or decreasing (−), as illustrated in Figure 1. Causal maps have been used for decades in Participatory Modeling (PM) to elicit the perspectives of various participants on a domain (Powell et al., 2024). On the one hand, the low requirements of causal mapping make it accessible across skill levels and settings (Knox et al., 2024), using paper notes or meeting rooms (e.g., moving sticky-notes and connecting them) or employing dedicated software such as Cmaps and Jmap (Jeong, 2016; Cañas et al., 2004). On the other hand, its structure

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suffices to support many subsequent analytical and modeling tasks. As emphasized in a dedicated volume, "a key reason for using causal maps [...] is the need to aggregate and comparatively analyze the knowledge and belief patterns of several individual actors and collectivities" (Laukkanen and Wang, 2016). Combining individual causal maps into a group-level map leverages the expertise and experiences of individuals to achieve a comprehensive understanding of a problem domain (Giabbanelli and Baniukiewicz, 2018), thus supporting tasks such as intervention design and evaluation (where should we apply a change in the system? what are the rippling consequences?). Analyzing individual maps reveals how people think about a problem (e.g., by identifying core concepts via centrality measures), which supports discussion between stakeholders groups by comparing perspectives, or a contrast between perceived mechanisms and real-world data. Causal maps also serve as precursors to Fuzzy Cognitive Maps (which have been used in over 20,000 studies per Kininmonth et al. (2021)), thus allowing modelers to perform numerical simulations to evaluate the level of change in a system in response to what-if scenarios[1]. For more examples of tasks involving causal maps, we refer the reader to Reddy et al. (2019).

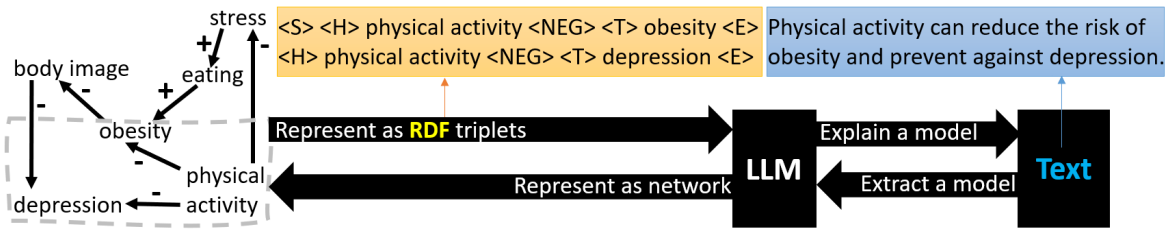

Fig. 1. A causal map (left) consists of labeled nodes and directed, typed edges for positive (+) and negative (−) causation. When the goal is to explain an existing map, the map or parts thereof are first expressed using the Resource Description Framework (RDF) as RDF triplets, then transformed into text. The RDF representation begins with the <S>*tart* tag, concludes with the <E>*nd* tag, and lists each edge as going from from a <H>*ead* node to a <T>*arget* node. Types are either <NEG>*ative* or <POS>*itive*. Conversely, if the objective is to structure a text as graph, key concepts are extracted and linked using a Question & Answer system based on the text. The LLM's knowledge model acts as a bridge in both cases.

Two limitations have been observed in the participatory modeling processes that create causal maps. First, the process is intuitive and transparent, as participants are guided in creating maps that they can see. However, when maps are integrated, the group-level map can become large and include several complex structures (e.g., multiple loops). One research group viewed such maps as an "almost incomprehensible web of interconnectedness [in which] the scale and number of interactions make it difficult to see how one might use it in any practical way to develop systemic approaches" (Siokou et al., 2014). Others deployed a culinary metaphor, considering that a large causal map "looks more like a spilled plate of spaghetti than anything of use" (Allender et al., 2015). Although several software have been developed specifically to visualize causal maps (often by tailoring methods by visualizing ontologies as covered in Dudáš et al. (2018)), usability studies are not used to interacting with networks even if their daily occupation involves systems thinking tasks (e.g., intervention design for public health) (Giabbanelli and Vesuvala, 2023). Second, participants may neither be available nor need to be involved if the goal is solely to build a map rather than engage participants throughout a process culminating in an intervention. For example, if a team wishes to map out the risks and protective factors to obesity, an alternative would be to read authoritative documents such as whitepapers and the peer-reviewed literature.

[1]Cognitive maps are related to causal maps, as they both have labeled nodes and directed edges. However, studies in cognitive mapping do not necessarily assign a sign to edges (Levy et al., 2018), which would be a requirement for causal maps. The related mapping approach of *influence diagrams* also involves labeled nodes and directed edges, but edges are unlabeled; *quantitative* influence diagrams assign a weight to each edge, but it is strictly positive (Moon and Adams, 2016).

These two limitations underpin the growing importance of *text* and Large Language Models (LLMs) in causal mapping (Figure 1). Instead of providing participants with an unwieldy network covering the entirety of their group's perspectives, an LLM can transform the map into a more universally understood format: text (Shrestha et al., 2022). Instead of asking questions to experts who would re-state what they have already written, an LLM can be used as part of a Question & Answer system to gradually build a map by asking questions to the text (Long et al., 2023; Giabbanelli and Witkowicz, 2024). As LLMs have recently been employed to support the tasks of map-to-text and text-to-map[2], it is important to provide guidance and reusable resources to nurture the development of this field.

Thus far, guidance has included surveys on causal inference via LLMs (Liu et al., 2024) or technical overviews of the use of LLMs in relation with conceptual models such as causal maps (Giabbanelli, 2023). This guidance promotes awareness of the potential and techniques necessary to build LLM-based tools for causal mapping, which could contribute to the growing development of such tools. At present, each research group uses their own datasets and assesses results using potentially different metrics. This was not an issue in the early exploration phase, which demonstrated technological feasibility. As we move to the next phase of research, a *comparison* of performances will gradually become necessary. Such a comparison requires some common datasets (i.e., *benchmarks*) and shared procedures for assessment. While the release of a causal dataset and associated codes is normally the subject of a dedicated work (Lam et al., 2024), our brief report gathers several datasets consisting of pairs of *(causal maps, text)* to support practitioners in performing assessments across different contexts (e.g., novice vs. expert writers, sentence generation vs. paragraph generation). In addition, we provide reusable code to support the map-to-text and text-to-map tasks. These resources are publicly available on a third-party repository (Open Science Framework) at https://osf.io/98kmy/ with a permissive license (MIT License) for re-use.

In summary, the resources provided and explained in this report are as follows:

- two datasets of several hundred entries, used for sentence-level generation and evaluation.
- three datasets consisting of entire causal maps and their accompanying narratives. We cover cases including novice writers, each producing three models from cases of their choice; fluent writers, each producing a model from a given case; and expert academic writers, who produced different models in peer-reviewed articles and textbooks. The diversity of these datasets supports paragraph-level generation and evaluation.
- a set of five notebooks (containing exercises and their solutions) to show step-by-step how to perform qualitative and quantitative assessments, with applications to both map-to-text and text-to-map

The remainder of this brief report is structured as follows. Section 2 explains how each dataset was collected and the type of data that it contains, along with brief statistical summaries. Section 3 details how the map-to-text and text-to-map can be assessed without ground truth data (i.e., based on expectations) or with the type of data that we have provided. Section 4 presents the notebooks that we created for this brief report, which demonstrate the use of the datasets and several assessment procedures.

2. Five Reusable Datasets to Benchmark Map-to-Text and Text-to-Map Across Domains

We collected five datasets to support a variety of tasks, writing styles, and application domains (Table 1). For each dataset, we provide (1) one comma-separated file consisting of a list of causal maps

[2]The task of text-to-map consists of creating a map from a corpus, based on the assistance of the knowledge model afforded by an LLM. While similar names are sometimes employed to align text with the best *available* model (Korel et al., 2023) and emerging software use LLMs to *expand* an existing map (Huang et al., 2023), we emphasize that our goal is to *build* the model.

using the RDF representation and the associated text; and (2) the causal maps as a comma-separated file consisting of a list of typed edges[3]. The youth **suicide** and **obesity** datasets *started with sizable causal maps*: the (large) suicide causal map totals 361 concept nodes and 946 causal edges, while the (medium-sized) obesity causal map has 98 nodes and 174 edges. Both maps were obtained from a participatory modeling process by aggregating the views of 15 (for suicide) and 19 academic experts (for obesity). This comprehensive level of details provides an interesting challenge as it can exceed the knowledge model of an LLM, thus they are best employed with either fine-tuning or in a few shot learning setting[4] rather than in a zero shot setting (Phatak et al., 2024). The maps were automatically divided into small subgraphs using a decomposition algorithm (Shrestha et al., 2022) and transformed into an RDF representation (Figure 1). The list of RDF representations were given to teams of students, who were asked to write the equivalent text. They could neither invent concepts or mechanisms nor omit any of the content, thus these datasets are designed for high coverage and faithfulness[5]. Three examples from the dataset on youth suicide are shown below.

<S> <H> Multi-generational despair <POS> <T> Acceptance of one's identity <H> Acceptance of one's identity <NEG> <T> Identity conflict <E>
⇒ *Experiencing multi-generational despair increases one's acceptance of their identity, so they experience lower rates of identity conflict.*
<S> <H> Separation <POS> <T> Being in foster care <E>
⇒ *A rise in separation of children from their parents leads to more children in foster care.*
<S> <H> Invalidation <POS> <T> Toxic stress <E>
⇒ *Feeling of invalidation enforces toxic stress.*

Table 1
Our five open datasets can support a variety of tasks in translating between causal maps and text.

Dataset	#pairs (map,text)	Granularity	Topics	Map source	Text source
Obesity	602	Sentence generation	Single	(Giabbanelli and MacEwan, 2024)	From map, by multiple students
Suicide	719			(Giabbanelli et al., 2022)	
Students	18	Paragraph generation	Multiple	Provided by writer	6 novice student writers
Lake	30		Single	Inferred from writer	30 advanced student writers
SD	20		Multiple	Published (various)	Expert academic writers

The other three datasets *associate paragraphs to each model* instead of operating at the sentence-level. They can thus be employed if the focus is not on fidelity with respect to fragments of the model, but rather on readability and flow of the overall text. The **students** dataset was collected by asking six novice writers trained in systems thinking to identify three complex problems of their choice, produce

[3]This standard file format can be directly loaded through a variety of libraries, such as Python's `NetworkX` via the `read_weighted_edgelist` function.

[4]An LLM is expected to provide satisfactory performances if the task/domain aligns with its training data. In addition, thanks to in-context learning and zero-shot capabilities, the knowledge models of LLMs can "generalize across tasks for which they were not explicitly trained" (Zhang et al., 2023). However, the models may need help when operating in highly specialized contexts. This can be as simple as providing a handful of samples (i.e., few shot learning) or it can involve more extensive fine-tuning, which must preserve the general linguistic ability of the LLM while improving its specialized functions.

[5]Faithfulness refers to accurately reflecting the facts provided in the causal map without any misrepresentation or injection in the text, whereas coverage refers to verbalizing all the details present in the map.

a causal map and narrate it. Problems included climate change, budget cuts for schools, disciplining a child, or colonizing Mars. The text description often starts with an introduction to the problem (e.g., "A big issue in the healthcare world is many patients coming in requesting antibiotics when in reality, they are suffering from a viral disease") and occasionally includes filler words (see short sample below). This dataset provides variability among writers (six writers) and within writers (three problems).

> National Security is a complex problem because it involves many people, groups, and lots of overlap. The movement is supposed to ensure the safety and comfort of people but often tends to be inconvenient and may be an abuse of power. This abuse causes a loss of basic civil rights, censorship, a greater cost, and an increased corporate involvement. As a whole, this could lead to increased terrorism threats and the unhappiness of the people. An increased national/governmental involvement is supposed to increase the safety of the people but to some people, this has merely taken away rights, made lives more complicated, and made it possible to use the people's fear.

The **lake** dataset comprises the responses of 30 participants to a vignette, Lake Urmia, depicting a multifaceted socio-environmental problem (Davis et al., 2020). Participants were graduate students recruited at Virginia Tech to answer to the vignette, and each received $5 Amazon gift card. The textual inputs from these individuals underwent individual analysis, employing independent coding by three system dynamics modelers who recorded the presence of links and loops (if present) in each response. Any disparities were addressed through collaborative case reviews. Subsequently, these codes were utilized to manually construct causal maps for each individual response. In sum, the dataset includes textual responses of the participants and corresponding maps generated by system dynamics experts. This dataset facilitates the assessment of LLM performance in capturing the mental models of non-experts.

Finally, the **System Dynamics** (SD) dataset includes 20 pairs of text and their corresponding maps presented in SD textbooks and journal articles authored by system dynamics experts (Hosseinichimeh et al., 2024). This dataset provides a sample to assess the accuracy of a LLM in reproducing the causal maps generated by a SD expert using the textual content extracted from these publications. We chose examples with relatively short descriptions of maps from SD publications. Some of the provided texts lack sufficient detail regarding the maps, with authors omitting explanations for certain components of feedback loops. Conversely, other texts offer comprehensive explanations of the loops. This diversity aids in evaluating a LLM's performance across a range of scenarios. To utilize the texts effectively as inputs, we exclude any references or utilization of the term 'feedback loop' to prevent a LLM from detecting the loop solely based on the term's presence. Additionally, if the map represents a generic model like the capability trap model and the text contains examples illustrating each generic variable, we omit these examples to maintain brevity and concentrate the text on the map itself.

The purpose and context of each dataset is reflected in the characteristics of its maps and texts (Table 2). Datasets for sentence-level generation consist of smaller fragments of a map, typically narrated in one or two sentences of less than 20 words. In contrast, datasets for paragraph-level generation consist of one or more paragraphs and complete maps that are either very focused (the Lake dataset has an average of 10 nodes and 6 edges) or more comprehensive (the Students dataset has 22 nodes and 20 edges on average). When using a LLM to convert between maps and text, we recommend starting with sentence-level datasets to ensure that the operations are working as intended. Then, a proposed solution can be scaled-up to paragraph-level datasets and compared with the rich descriptions produced by a variety of writers (novice, graduate students, experts), depending on the target audience relevant to a project.

Table 2

For each dataset, values are reported as mean ± standard deviation (minimum—maximum).

Dataset	Characteristics of causal maps		Characteristics of texts	
	#nodes	#edges	#sentences	#words
Obesity	5.257 ± 3.947 [2—26]	3.355 ± 3.182 [1—23]	1.473 ± 0.797 [1—7]	19.716 ± 13.111 [4—72]
Suicide	3.951 ± 2.419 [2—18]	2.189 ± 1.498 [1—9]	1.282 ± 0.646 [1—6]	16.122 ± 9.276 [4—60]
Students	21.889 ± 5.614 [14—35]	19.556 ± 11.858 [10—53]	9.389 ± 4.118 [3—18]	186.333 ± 116.757 [49—465]
Lake	9.767 ± 5.217 [2—21]	6.433 ± 3.401 [1—14]	7.667 ± 4.237 [2—23]	146.533 ± 85.759 [28—426]
SD	17.950 ± 8.513 [6—34]	12.300 ± 6.490 [4—23]	7.000 ± 5.657 [1—26]	135.500 ± 102.933 [19—470]

3. Assessment Techniques: Established and Potential Metrics for Map-to-Text and Text-to-Map

Although we focus on automatic evaluation, human evaluation is necessary to complement automated methods. Automatic evaluation can be broadly categorized along the lines of *reference-based* (where model outputs are evaluated by comparing them to human gold truth reference answers) or *reference-free* (when human gold truth is unavailable) methods. Both types of metrics have their own strengths and weaknesses. For instance, reference-based metrics provide a clear benchmark for comparison by using human-generated gold-standard references and form the basis of several well-established metrics (e.g., BLEU, ROUGE, METEOR). However, they also require high-quality references which can be time-consuming and costly to obtain, may not capture the full range of acceptable outputs as there can be multiple valid ways to express the same information, and may not account for the inherent variability in human language generation. Reference-free metrics, on the other hand, do not require human-generated references, making them scalable, cost-effective, and more tolerant of linguistic diversity. However, reference-free metrics are relatively newer and less established, with ongoing research on their robustness and correlation with human judgments. Although some studies have found reference-free metrics to obtain higher correlation with human annotations as compared to reference-based metrics (Gowda et al., 2023), they can be susceptible to spurious correlations (Durmus et al., 2022; Deutsch et al., 2022). Ridenour et al. (2022) found that reference-based and reference-free metrics capture different aspects of quality, and recommended that both should be reported to provide a comprehensive assessment of the model performance.

3.1. From Text to Causal Map

First, we discuss the referent-based evaluation of causal maps generated from text where we compare the model output (linearized representation of a set of triples including the head entity, tail entity and the causal relationship between the two entities) to a human-created gold standard causal map containing the correct entities and relationships. Metrics like precision, recall, and F1 score[6] are typically used to assess how well the generated output captures the essential elements of the causal map (Mihindukulasooriya et al., 2023). These metrics can be used in two ways for evaluating automatically generated graphs: node-level evaluation (considering only the entities; edges are not considered) and edge-level evaluation (considering only the relationships; entities are not considered). Additionally, there is a vast literature on assessing novice maps (i.e., produced by students) in educational technology (Figure 2), which could be

[6]Precision measures the proportion of correctly identified triples (entities and relationships) in the generated graph compared to the human gold truth. Recall measures the proportion of relevant triples in the gold truth that are correctly identified in the generated graph. F1 score, which is the harmonic mean of precision and recall, provides a balanced performance measure.

re-purposed in the future to assist with the assessment of maps generated by LLMs. From a *referent-free* perspective, the objective in educational technology is to examine whether a student produced a good map with respect to set criteria (e.g., sparse, connected, with at least one loop). Although there are at least 12 potential criteria to define a 'good map', several of them are correlated and the choice of criteria also depends on how the map was produced (e.g., the diameter is useful to score a causal map produced by analyzing a report) (Wang and Giabbanelli, 2023). When taking a *referent-based* approach, metrics can be grouped in four categories (Figure 2): embeddings, kernels, Graph Edit Distance (GED) (Gao et al., 2010), or the use of a single measure. The GED holds significant potential because it identifies the series of steps necessary to transform one map (e.g., generated by LLM) into a target map (e.g., the ground truth). It could thus serve both as a scoring method and a guidance tool to improve the transformation.

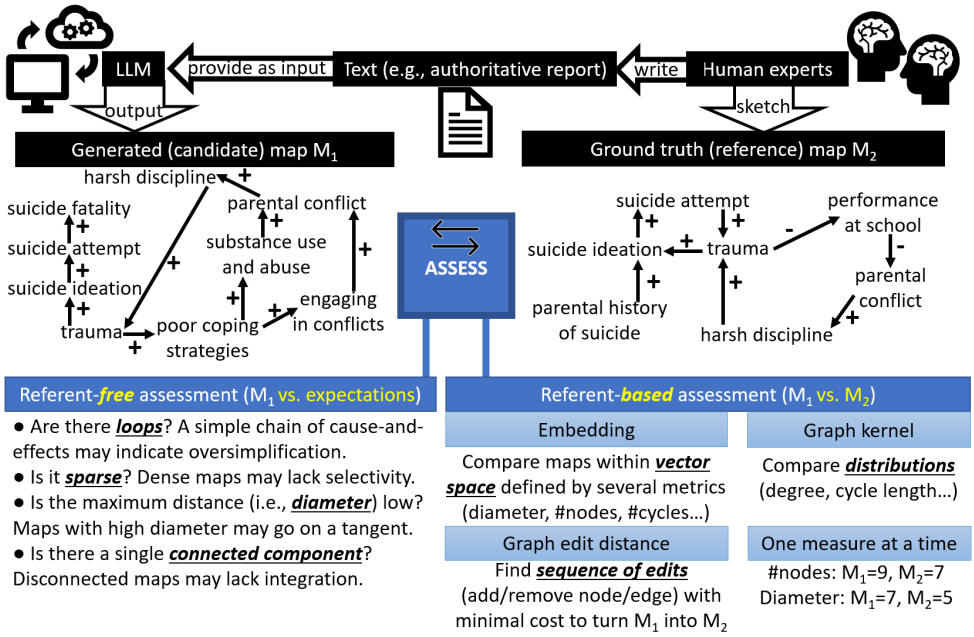

Fig. 2. The literature on map assessment from educational technology has compared the maps of 'novice' learners with 'expert' maps, or suggested several measures that would correlate with quality in a novice map.

The entities in the generated maps and the ground truth maps are likely to have variation in their terminology (e.g., suicide fatality vs. death by suicide), thus it may be helpful to also report metrics that measure the degree of relevant matching using metrics derived from semantic similarity. Because triples are likely to contain words rather than full sentences, word embeddings such as `word2vec` and `GloVe` that provide high-dimensional vectors representing the meaning of a word in relation to other words can be used to assess semantic similarity. Words with similar meanings tend to have embeddings that are closer together in this vector space. The distance between these embeddings can be measured using various metrics but a common one is cosine similarity which ranges from 0 (completely dissimilar) to 1 (identical). However, such word embedding based metrics are less suitable for evaluating the causal relationships as they often fail to capture nuanced meaning when comparing synonyms[7]. Therefore,

[7] `word2vec` produces a higher similarity score between antonyms such as 'increase' and 'decrease' (0.837) than it does between 'increase' and its synonyms such as 'rise' (0.635), 'grow' (0.410), or 'amplify' (0.192).

for evaluating the causal relationships, it is useful to report simpler n-gram metrics such as BLEU and ROUGE as they focus on exact word matching, crucial for causal relationships.

3.2. From Causal Map to Text

Typical metrics comparing the generated text to one or more reference texts written by humans include BLEU, ROUGE and CIDEr. However, because these metrics focus on identifying *n*-grams (sequences of *n* words) that match between the generated and reference texts, they often fail to capture the diversity in natural language generation. METEOR addresses some of the shortcomings of popular metrics like BLEU by focusing on identifying semantically equivalent or stemmed versions of words rather than strict word matching, thereby allowing for more flexibility and capturing paraphrasing. However, it relies on external resources like synonym databases and is computationally more expensive compared to simpler metrics. Instead of relying on word-to-word overlap, metrics such as BERTScore that focus on semantic similarity between generated and reference texts can capture more nuanced semantic relationships between texts and allow for flexibility in how information is expressed. BERTScore utilizes word embeddings from pretrained language models such as BERT and calculates the cosine similarity between the embedding vectors of the generated and reference texts.

Instead of semantic similarity, one can also rely on semantic inference measures which have been found effective in identifying faithfulness (Maynez et al., 2020). Natural language inference (NLI) is a task where given two pieces of text – a premise (LLM's output) and a hypothesis (human gold truth reference text) – the goal is to determine whether the hypothesis can be inferred from the premise, with common relationships including entailment, contradiction, and neutral. Several benchmarks for evaluation via NLI models (e.g., ANLI, SummaC, or FactCC) can be leveraged or a new model can be fine-tuned for this specific task to determine whether a given text is consistent or inconsistent with the source text (Soleimani et al., 2023). Because assessing faithfulness requires logical inference over factual information, several question answering based models can also be used, such as QAGS (Wang et al., 2020) and FEQA (Durmus et al., 2020). Another metric commonly used is PARENT (Dhingra et al., 2019) which uses not only the reference text but also the structured or semi-structed data (such as tables or maps) when evaluating the generated text (Tian et al., 2019). LLM-based evaluators are an emerging research area with potential for large-scale, adaptable assessments (Gao et al., 2024). However, a challenge is avoiding 'conflicts of interest' by using the same LLM for both generation and evaluation.

4. Reusable Scripts to Perform and Evaluate Map-to-Text and Text-to-Map Transformations

In section 2, we summarized five datasets with different characteristics. In section 3, we discussed how the tasks of map-to-text and text-to-map can be assessed. To fully support users in exploring our datasets and applying the notions aforementioned, we have developed a set of five Jupyter Notebooks for this report. Each notebook comes in two versions: one with exercises and an empty space to perform them (thus encouraging users to acquire skills through practice), and another one with all exercises completed (thus allowing users to follow a demonstration). The notebooks were designed as a set through which users can progress, starting with discovering the API to interact with a LLM (notebook #1), generating text from maps (notebook #2), evaluating it manually (notebook #3), optimizing the process (notebook #4), and building a complete text-to-map system (notebook #5). The notebooks were internally evaluated as part of an onboarding process in our group and iteratively revised for clarity based on the feedback.

Table 3
Our five open Jupyter Notebooks teach the core concepts necessary to support text-to-map and map-to-text tasks.

Notebook	Overall theme	Specific aspects covered
1	Using the OpenAI API for GPT	Choosing a model, setting parameters, defining role, using styles.
2	Performing/assessing map-to-text	Automatic sentence-level metrics (e.g., BLEU, ROUGE) and paragraph-level metrics (Flesch readability).
3	Manual assessment for map-to-text	Manual evaluation via coverage, fluency, faithfulness on a 5-point Likert scale.
4	Optimization	Assessing the effect of parameters/models onto map-to-text metrics. Using pairs of text-map for training (fine-tuning, zero-shot, few-shot).
5	Performing text-to-map	Build a Question&Answer (Q&A) system to turn a PDF document into a map.

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